

Theoretical Concepts of Machine Learning

Exam 1

April 11, 2019

- What is the mathematical definition of *Risk*? (4)
- How do you estimate Risk for (6)
 - big data-sets?
 - small data-sets?
 - which assumptions and problems are there for small data-sets?
- What is a *generative model*? (2)
- What is the *Fisher information matrix*? (4)
- What is the *Cramer-Rao Lower Bound* (formula)? (8)
- What is an *unbiased Estimator*? (4)
- Name the properties of the *Maximum Likelihood Estimator (MLE)* (8)
- What is *Expectation Maximisation*? (4)